

Liquid YPD Preparation

Liquid medium allows for rapid growth and easy measurement via OD, and YPD provides the nutrients necessary for auxotrophic strains to grow [7]. This protocol makes 1L of liquid yeast peptone dextrose medium (YPD) at 1% yeast extract, 2% peptone, and 2% dextrose concentration.

Reagents

✓	Name	Quantity	Location/Notes
☐	Peptone	20g	
☐	Yeast Extract	10g	
☐	ddH ₂ O	1L	
☐	D-glucose / dextrose	20g	
☐	Sterile water	As needed	

Equipment

✓	Name	Quantity	Location/Notes
☐	Analytical scale		
☐	Autoclave		Training required
☐	Stir bar		
☐	500mL autoclaveable bottle		Used to make dextrose

		solution	
☐	1L autoclaveable bottle		Used to store media
☐	BSC		Training required

Instructions

1. Prepare the dextrose solution by adding the following to the **500mL bottle**, use a stir bar to stir:

Reagent Tracker	Quantity
☐ D-glucose / dextrose	20g
☐ ddH ₂ O	to 200mL

2. Prepare the rest of the solution by adding the following to the **1L bottle**, use a stir bar to stir:

Reagent Tracker	Quantity
☐ Peptone	20g
☐ Yeast Extract	10g
☐ ddH ₂ O	to 800 µl

-
3. Autoclave both liquids at 121°C for 15min [3] and allow them to cool
 4. In a BSC, add the dextrose solution to the 1L bottle, add sterile water until the solution is 1L

Notes here

1. Can be stored in 4°C fridge for a few months [5] [6]
2. DO NOT RE-AUTOCLAVE (see below)
3. Solutions must be autoclaved separately and later combined to avoid triggering the maillard reaction in the autoclave , which could make toxic by-products and lower glucose concentration [4]

References

- [1] <https://www.sciencedirect.com/science/article/pii/S1359511322003786>
- [2] <https://www.protocols.io/view/yeast-peptone-dextrose-ypd-medium-x54v9d7xl93e/v1>
- [3] <https://www.sigmaaldrich.com/deepweb/assets/sigmaaldrich/product/documents/105/767/y1500pis-ms.pdf>
- [4] https://www.researchgate.net/post/Autoclaving_Glucose_in_yeast_peptone_Dextrose_plate_with_Agar
- [5] <https://www.researchgate.net/post/How-long-can-I-store-un-poured-LB-YPD-agar-for-plates-in-duran-bottles>
- [6] <https://shop.gmexpression.com/YPD-medium-liquid>
- [7] https://www.researchgate.net/publication/311780777_Isolation_and_Characterization_of_a_Variant_Manganese_Resistant_Strain_of_Saccharomyces_cerevisiae